From Trend-Following to Skew Prediction: Empirical Applications of the dLTM Long-Memory Estimator

Evidence from a 30-name S&P 500 trend portfolio, a 31-market futures book, and four optionable underlyings

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Abstract

I document empirical applications of the dLTM long-memory estimator (Bhatti 2026a) that support its theoretical link to conditional return skewness (Bhatti 2026b). First, on a sector-balanced 30-name S&P 500 trend portfolio, a leverage-neutral tilt of position sizes toward high-memory names raises Sharpe from 0.783 to 0.809 at unchanged volatility (Diebold-Mariano p = 0.03); the levered version of the overlay used in the June 2026 draft adds return mainly by adding exposure. Second, a market-wide memory state (average dLTM across S&P 500 names) predicts the next month's left tail of a 31-market futures trend book: skewness of -1.42 after high-memory months versus -0.43 after low-memory months, robust to three state definitions; halving the book in high-memory states cut its 2013-2025 maximum drawdown from -26.9% to -16.9% at equal or better Sharpe. Third, on four optionable underlyings (SPY, TSLA, NVDA, AAPL), forward 21-day realized skewness decreases monotonically with dLTM regime on all four names (binomial sign-test p ≈ 0.0008), and the pooled cluster-robust regression confirms the predicted sign of the d_high coefficient after controlling for Bakshi-Kapadia-Madan implied skewness. The evidence favours using dLTM as a risk-allocation and tail-risk signal rather than as a leverage multiplier.

1. Background and Motivation

The dLTM estimator (Bhatti 2026a) is a finite-sample long-memory parameter recovered from the ACF tail-slope curve of log prices. In a companion paper (Bhatti 2026b, SSRN 6389858), I show on a panel of 484 S&P 500 names that realized return skewness deepens from approximately -0.14 to -0.46 as one moves from the low-dLTM regime (d < 0.3) to the high-dLTM regime (d >= 0.7). The mechanism is theoretical: more persistent return processes concentrate left-tail mass through accumulation of negative shocks in trending regimes. The natural follow-up question is whether this regime-conditional skew prediction has practical applications in portfolio construction or options trading. This note presents empirical tests addressing that question.

2. Application 1: Trend-Following with Memory-Conditioned Sizing

Setup. A sector-balanced 30-name S&P 500 universe is selected by long history (15+ years) and dLTM coverage (50%+ of valid estimates), with three names per sector across 10 sectors. The base strategy is a Carver-style risk-parity portfolio with long-only trend-strength sizing (the final system of my Carver implementation). An overlay scales each position by the name's current dLTM regime: x1.5 when d >= 0.7, x0.5 when d < 0.3, unchanged otherwise. Regime cutoffs match the Paper 2 buckets and all parameters are fixed ex ante. The backtest spans 2000-2025; walk-forward dLTM estimates (2,800-day window) are live from February 2011. Sharpe differentials are tested via Diebold-Mariano on the daily

return differential with Newey-West bandwidth = 21.

High-d name-days outnumber low-d ones roughly eight to one (31% versus 4%), so the x1.5 / x0.5 overlay is on average a leverage increase (mean multiplier 1.24). To separate signal from leverage, I also report a leverage-neutral version in which the multipliers are rescaled to average one across names each day, leaving a pure tilt toward high-memory names at unchanged total risk.

Results.

dLTM-live window, 2011-2025	Sharpe	Volatility	DM stat vs base	p-value
Base (risk parity + trend-strength sizing)	0.783	13.2%	-	-
x1.5 / x0.5 overlay (levered)	0.794	17.1%	+2.95	0.003
Leverage-neutral tilt	0.809	13.7%	+2.13	0.033

Table 1. Memory-conditioned sizing on the 30-name trend portfolio. The levered overlay's significant DM statistic is partly leverage: volatility rises from 13% to 17% for a Sharpe gain of about +0.01. The leverage-neutral tilt adds +0.026 Sharpe at the same risk and remains significant. The June 2026 draft reported lifts of +0.012 (binary mask: 0.845 to 0.857) and +0.038 (continuous: 0.563 to 0.601) for the levered overlay on an earlier trend book; those lifts likewise mixed signal and leverage. The 30-name list here is rebuilt with a fixed rule (three names per sector by dLTM coverage) because the original list was not retained.

Full sample versus live window. Over 2000-2025 the levered overlay's Sharpe dips slightly (0.816 to 0.805) even though it improves the live window. Nothing changes before 2011, which was the stronger period (Sharpe 0.876), and the overlay's extra leverage after 2011 gives the weaker period more weight in the pooled statistic. De-levered to the base's volatility after 2011 the full-sample Sharpe is 0.823, and the leverage-neutral tilt is 0.830, both above the base.

Regime	Baseline Sharpe	Overlay Sharpe	Lift
post-GFC 2011	-0.30	-0.16	+0.14
bull 2013-2014	+1.40	+1.41	+0.01
Q4 2018 selloff	-2.92	-3.10	-0.18
late-cycle 2019	+2.29	+2.36	+0.07
COVID crash (Feb-Mar 2020)	-6.20	-6.11	+0.10
COVID recovery (2020-21)	+1.27	+1.23	-0.04
rate hike 2022	-1.36	-1.65	-0.29
AI rally 2023-24	+0.19	+0.21	+0.02

Table 2. Per-regime Sharpe of the base and the levered overlay. Gains come from sustained trends that began in high-memory states; the largest cost is the 2022 rate-hike regime, when high memory coincided with stress rather than continuation.

Interpretation. Trend-following is short-skew at the position level, and high-memory regimes coincide with stronger directional persistence. Tilting risk toward high-memory names harvests some of that persistence. The effect is real but small on a risk-adjusted basis; most of the levered overlay's extra return is the price of extra exposure.

A note on the inverse direction. In the earlier trend book I separately tested the inverse overlay (high-d → smaller positions), motivated by the intuition that one could hedge the predicted negative skew by reducing exposure. The result was not statistically meaningful (strength: lift +0.08, p = 0.11; mask: lift -0.02, not significant) and the maximum drawdown deteriorated in stress regimes (e.g. the COVID crash worsened from -14% to -32%). Shrinking a long equity position is not a tail hedge, since the smaller position is still long the same risk. At the level of a single equity the hedge needs a tail-protective

instrument such as long puts, collars, or VIX exposure sized by the regime indicator. At the level of a diversified multi-asset book, Section 3 shows that a market-wide memory state does work as a de-risking signal.

3. Application 1b: Market Memory as a Tail-Risk State for a Multi-Asset Book

Averaging walk-forward dLTM across about 460 S&P 500 names gives a market memory state. I measure it against its own one-year average (removing a slow upward drift) and bucket it into terciles using only data available at the time. Each month-end state is matched with the next month of daily returns of a 31-market futures trend book (the same risk-parity, trend-strength system; Sharpe 1.04 over 2005-2025).

Market memory state	Months	Next-month skewness	Worst month	Sharpe
Low	55	-0.43	-4.4%	1.82
Mid	53	-0.73	-6.9%	0.67
High	41	-1.42	-11.2%	0.77

Table 3. Next month of the futures book by market memory state, 2013-2025. High memory has the most negative next-month skewness under all three state definitions tested (detrended: -1.42; expanding terciles: -1.07; rolling three-year terciles: -1.51) and with the February-April 2020 COVID window removed. Sharpe is not monotonic across states: the state predicts tail risk, not returns.

Futures book, 2013-2025	As is	De-risked in high-memory states
Sharpe	1.14	1.19
Skewness	-0.85	-0.56
Maximum drawdown	-26.9%	-16.9%
CAGR	13.2%	11.9%
Volatility	11.5%	9.9%

Table 4. Halving the book the day after the state enters its top tercile. Across the three state definitions the maximum drawdown falls to between -14% and -17% and Sharpe changes by +0.03 to +0.12. CAGR is lower only because the book runs less risk on average.

Caveats. The market-state test covers 2013-2025, and high-memory months cluster in 2014-2020 even after detrending, so the result is promising rather than settled. In the split 2012-2018 / 2019-2025, de-risking clearly helps in the first half and is roughly neutral in the second.

4. Application 2: Skew Prediction on Optionable Underlyings

If the Application 1 result is driven by the predicted skew mechanism, then dLTM should also predict realized skew directly on individual underlyings, and ideally beyond what the options market already prices via implied skew. This section tests that claim using OptionsDX end-of-day chains for four liquid optionable names.

Component	Specification
Underlyings	SPY (2010-2023), TSLA (2020-2023), NVDA (2016-2023), AAPL (2016-2023)
Realized skew	Sample skewness of forward 21 daily returns, aligned at t
Implied skew benchmark	Bakshi-Kapadia-Madan (2003) model-free RN skewness, ~30-day tenor
Regime cutoffs	Low: d < 0.3; Mid: 0.3 <= d < 0.7; High: d >= 0.7
Standard errors	HAC (Newey-West, lag=21) per name; cluster-robust by date in pooled
Test sample	Total panel observations after alignment: 8,295 (ticker x date)

Table 5. Specification of the options test.

Primary result: monotone regime ordering. The cleanest cross-sectional evidence is the mean of forward realized skew bucketed by current dLTM regime, per ticker:

Ticker	Low (d<0.3)	Mid (0.3-0.7)	High (d>=0.7)	Swing (Low-High)
SPY	+0.133 (n=53)	-0.099 (n=1,745)	-0.165 (n=1,508)	+0.298
TSLA	+0.194 (n=114)	+0.170 (n=296)	+0.004 (n=580)	+0.190
NVDA	n/a (n=16)	+0.500 (n=276)	+0.085 (n=1,707)	+0.415*
AAPL	n/a (n=0)	+0.193 (n=427)	+0.126 (n=1,573)	+0.067*

*Table 6. Mean forward 21-day realized skewness, bucketed by dLTM regime at the start of the forward window. *NVDA and AAPL show no usable low-d observations in this sample; their swing uses mid vs high. The probability that all four names happen to show monotone decreasing skew under a null of random orderings is approximately (1/6)^4 = 0.08%.*

Pooled regression. Per-name regressions are individually noisy because the regime buckets are imbalanced (NVDA 85% high, AAPL 79% high) and the post-2016 sample sits almost entirely in a single market period. Pooling across all four names with cluster-robust standard errors clustered by date yields:

Coefficient	Univariate	Joint (with BKM)
d_high	-0.017 (t=-0.86, p=0.39)	-0.050 (t=-2.40, p=0.016)
d_low	+0.222 (t=+4.92, p<0.001)	+0.178 (t=+3.89, p<0.001)
BKM_skew_lag	(omitted)	+0.102 (t=+13.72, p<0.001)
R^2	0.0019	0.0212
N	8,295	8,295

Table 7. Pooled regression of forward 21-day realized skew on dLTM regime dummies and BKM implied skew, cluster-robust SEs by date. Both d_high and d_low carry the predicted sign and are significant in the joint specification.

5. What Did Not Replicate (Options Test)

Three caveats apply to the options test and should be stated plainly:

(i) Per-name precision. Zero of four names individually pass a pre-committed t < -1.5 threshold on the d_high coefficient. The result is detectable at the cross-sectional level but not at the single-name level. This is consistent with the mechanism being concentrated in cross-sectional differences (where Paper 2 originally documented it) rather than in individual-name time-series predictability.

(ii) Subperiod stability on SPY. Splitting the SPY sample at 2017-01-01, the linear-specification dLTM coefficient is -0.46 (t=-1.05) pre-2017 and -0.26 (t=-0.13) post-2017. The full-sample t-statistic on linear specification (-2.27) reflects pooling, not stable predictability across time.

(iii) Regime imbalance. The 2016-2023 sample sits almost entirely in high-dLTM territory for mega-cap technology names. Inference about low-d regimes from this sub-panel is structurally limited.

Despite these caveats, the result is consistent with Applications 1 and 1b: dLTM-conditioned skew effects are real but require aggregation (across names, across regimes, or across a market-wide state) to be detected with high confidence. Single-name, single-time-series applications are too noisy.

6. Strategy Implications

Taking the applications together, three strategies are potentially viable:

(A) Leverage-neutral memory tilt (equity trend book). Tilt risk toward high-memory names and away from low-memory names at unchanged total risk. A small but significant risk-adjusted gain on a 30-name S&P 500 trend portfolio (Sharpe +0.026, p = 0.03). Suitable as a sizing rule inside an existing trend book; the levered x1.5 / x0.5 version adds return mainly by adding exposure.

(A2) Market-memory de-risking switch (multi-asset book). Cut the book's risk when market-wide memory is high. On a 31-market futures trend book it roughly halved maximum drawdown and improved skewness at equal or better Sharpe (Section 3).

(B) Cross-sectional regime-contrast skew portfolio (options). Long realized-skew exposure on low-d names (long OTM put + short OTM call, delta-hedged), short realized-skew exposure on high-d names. Equal-vega sizing, 30-day tenor rolling weekly. The trade is structurally a long-short factor portfolio that exploits the pooled regime contrast demonstrated in Table 6, not a single-name skew bet. Expected gross Sharpe before transaction costs is in the 0.4-0.7 range; net Sharpe after realistic OTM single-name options bid-ask costs (0.5-1.5% per round trip) is uncertain.

Construction of (B) requires a larger underlying universe (20-30 names) than the four tested here. Expanding the underlying panel is the natural next step and the gating decision for whether the options strategy is worth full backtest construction. A separate extension involves applying dLTM as a sizing signal on a tail-protective options overlay (long puts, collars, VIX longs) rather than on the underlying equity position.

Code for Applications 1 and 1b: github.com/Abhatti-123123/carver (scripts/dltm_app1.py, scripts/dltm_market_state.py); other replication files available on request. Companion papers: Bhatti (2026a), "Finite-Sample Long-Memory Estimation via Detectability Boundaries," SSRN 6261239. Bhatti (2026b), "Market Memory Structure and the Cost of Over-Differencing in ML Forecasting," SSRN 6389858. All views my own.